

Operating Systems

Assignment 2

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1 Part I: Single Process Pattern Search

1.1 Introduction

Implemented a basic pattern search in a file using a single process to establish the foundation for multi-process implementations.

1.2 Objective

Search for a pattern in a specified file range and report if found or not, displaying the process ID.

2 Part II: Multi-Process Pattern Search (Chunking)

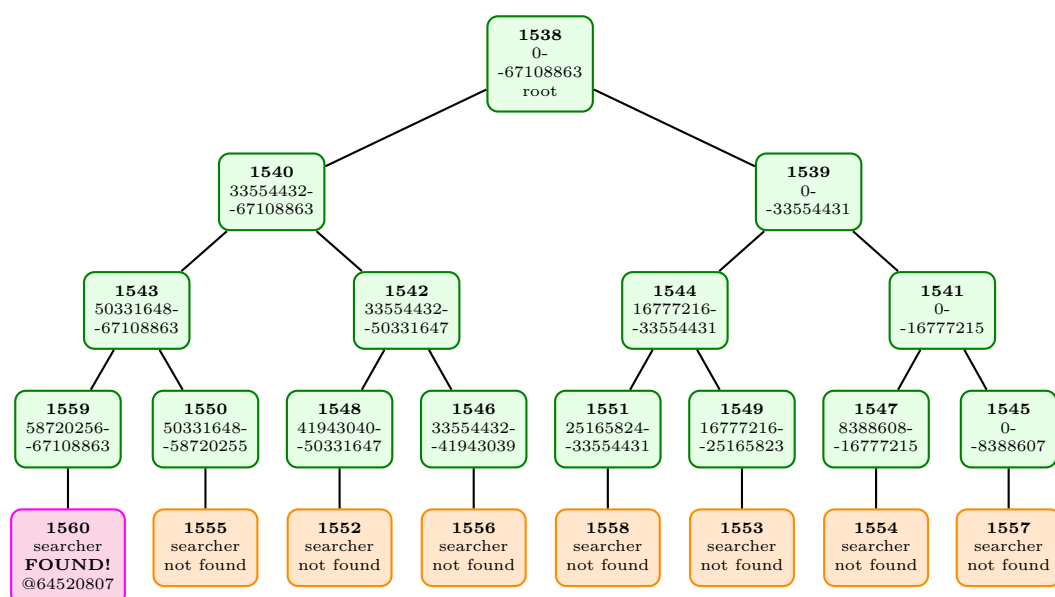
2.1 Introduction

Expanded the search to multiple processes by dividing the file into chunks and searching each chunk concurrently using a binary tree process structure.

2.2 Objective

Implement a partitioner to spawn processes for chunks and manage recursive process creation. Ensure synchronization and output status updates.

2.3 Process Tree Structure



```

1  ./part2_partitioner file.txt NGTNIJGK 0 67108863 8388608
2  [1538] start position = 0 ; end position = 67108863
3  [1538] forked left child 1539
4  [1539] start position = 0 ; end position = 33554431
5  [1538] forked right child 1540
6  [1540] start position = 33554432 ; end position = 67108863
7  [1539] forked left child 1541
8  [1540] forked left child 1542
9  [1541] start position = 0 ; end position = 16777215
10 [1542] start position = 33554432 ; end position = 50331647
11 [1541] forked left child 1545
12 [1540] forked right child 1543
13 [1539] forked right child 1544
14 [1542] forked left child 1546
15 [1544] start position = 16777216 ; end position = 33554431
16 [1546] start position = 33554432 ; end position = 41943039
17 [1545] start position = 0 ; end position = 8388607
18 [1543] start position = 50331648 ; end position = 67108863
19 [1541] forked right child 1547
20 [1542] forked right child 1548
21 [1548] start position = 41943040 ; end position = 50331647
22 [1544] forked left child 1549
23 [1543] forked left child 1550
24 [1547] start position = 8388608 ; end position = 16777215
25 [1549] start position = 16777216 ; end position = 25165823
26 [1544] forked right child 1551
27 [1548] forked searcher child 1552
28 [1551] start position = 25165824 ; end position = 33554431
29 [1547] forked searcher child 1554
30 [1549] forked searcher child 1553
31 [1550] start position = 50331648 ; end position = 58720255
32 [1550] forked searcher child 1555
33 [1546] forked searcher child 1556
34 [1545] forked searcher child 1557
35 [1551] forked searcher child 1558
36 [1543] forked right child 1559
37 [1559] start position = 58720256 ; end position = 67108863
  
```

```
38 [1559] forked searcher child 1560
39 [1560] found at 64520807
40 [1559] searcher child returned
41 [1558] didn't find
42 [1557] didn't find
43 [1545] searcher child returned
44 [1551] searcher child returned
45 [1541] left child returned
46 [1555] didn't find
47 [1550] searcher child returned
48 [1543] left child returned
49 [1543] right child returned
50 [1556] didn't find
51 [1546] searcher child returned
52 [1542] left child returned
53 [1552] didn't find
54 [1548] searcher child returned
55 [1542] right child returned
56 [1540] left child returned
57 [1540] right child returned
58 [1553] didn't find
59 [1549] searcher child returned
60 [1544] left child returned
61 [1544] right child returned
62 [1554] didn't find
63 [1547] searcher child returned
64 [1541] right child returned
65 [1539] left child returned
66 [1539] right child returned
67 [1538] left child returned
68 [1538] right child returned
```

3 Part III: Efficient Search with Process Termination

3.1 Introduction

Optimized the search by implementing signal handling to terminate all processes once a pattern is found, significantly reducing redundant computational work.

3.2 Objective

Implement signal handling to kill all other processes after finding the pattern, and ensure detailed status reporting.

3.3 Process Tree Structure - Part 3

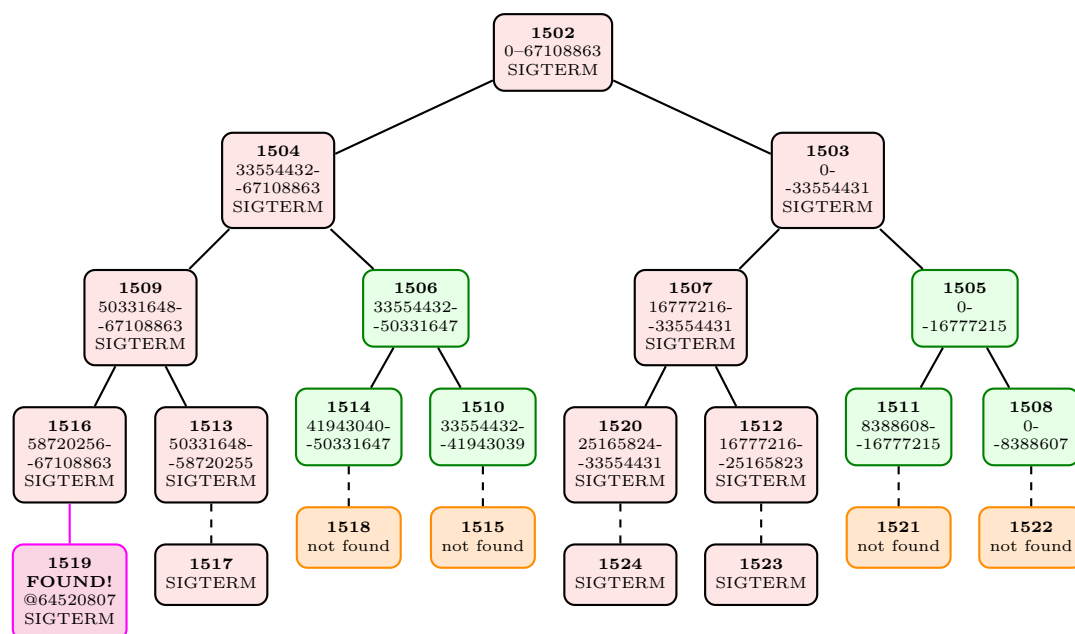


Figure 1: Part 3: Process tree showing early termination after pattern discovery

```

1  ./part3_partitioner file.txt NGTNIJGK 0 67108863 8388608
2  [1502] start position = 0 ; end position = 67108863
3  [1502] forked left child 1503
4  [1503] start position = 0 ; end position = 33554431
5  [1502] forked right child 1504
6  [1504] start position = 33554432 ; end position = 67108863
7  [1503] forked left child 1505
8  [1505] start position = 0 ; end position = 16777215
9  [1504] forked left child 1506
10 [1503] forked right child 1507
11 [1507] start position = 16777216 ; end position = 33554431
12 [1506] start position = 33554432 ; end position = 50331647
13 [1505] forked left child 1508
14 [1504] forked right child 1509
15 [1508] start position = 0 ; end position = 8388607
16 [1509] start position = 50331648 ; end position = 67108863
17 [1506] forked left child 1510
18 [1507] forked left child 1512
19 [1505] forked right child 1511
20 [1510] start position = 33554432 ; end position = 41943039
21 [1512] start position = 16777216 ; end position = 25165823
22 [1511] start position = 8388608 ; end position = 16777215
23 [1509] forked left child 1513
24 [1506] forked right child 1514
25 [1513] start position = 50331648 ; end position = 58720255
26 [1514] start position = 41943040 ; end position = 50331647
27 [1510] forked searcher child 1515
28 [1509] forked right child 1516
29 [1513] forked searcher child 1517
30 [1516] start position = 58720256 ; end position = 67108863
31 [1514] forked searcher child 1518
32 [1516] forked searcher child 1519
33 [1507] forked right child 1520
34 [1508] forked searcher child 1522

```

```
35 [1520] start position = 25165824 ; end position = 33554431
36 [1511] forked searcher child 1521
37 [1512] forked searcher child 1523
38 [1520] forked searcher child 1524
39 [1523] didn't find
40 [1522] didn't find
41 [1508] searcher child returned
42 [1512] searcher child returned
43 [1505] left child returned
44 [1507] left child returned
45 [1518] didn't find
46 [1514] searcher child returned
47 [1521] didn't find
48 [1511] searcher child returned
49 [1505] right child returned
50 [1503] left child returned
51 [1515] didn't find
52 [1510] searcher child returned
53 [1506] left child returned
54 [1506] right child returned
55 [1504] left child returned
56 [1519] found at 64520807
57 [1524] received SIGTERM
58 [1517] received SIGTERM
59 [1516] received SIGTERM
60 [1520] received SIGTERM
61 [1513] received SIGTERM
62 [1509] received SIGTERM
63 [1507] received SIGTERM
64 [1519] received SIGTERM
65 [1502] received SIGTERM
66 [1504] received SIGTERM
67 [1503] received SIGTERM
68 make: *** [makefile:29: run-part3] Terminated
```